

Beyond the Hype: A Rigorous Systems, Statistical, and Epistemological Audit of Self-Consistency, LLM-as-a-Judge, and Multi-Agent Clinical Architectures

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Abstract

Systematic Literature Review on Self-Consistency, LLM-as-a-Judge, and Multi-Agent Systems.

1 Abstract

As large language models (LLMs) transition from static, single-pass generation toward dynamic, multi-path reasoning and automated evaluation frameworks, the field of artificial intelligence faces a critical convergence of systems engineering bottlenecks and statistical validation deficits. This literature review provides a rigorous, multi-disciplinary audit of three foundational paradigms: (1) multi-path decoding via Self-Consistency, (2) automated model evaluation using the LLM-as-a-Judge framework, and (3) specialized clinical multi-agent architectures, represented by Cardiology-Chat.

Through a systematic synthesis of [Wang et al.(2022)Wang, Wei, Schuurmans, Le, Chi, Narang, Chowdhery, and Zhou] (and its peer-reviewed counterpart [Author(2023)]), [Gu et al.(2024)Gu, Jiang, Shi, Tan, Zhai, Xu, Li, Shen, Ma, Liu, Wang, Z], and [Z et al.(2026)Z, C, SS, X, Y, and Q.], we expose critical methodological vulnerabilities that threaten the scientific validity of these advancements. Specifically, we deconstruct the "compute-equivalent baseline deficit" in multi-path decoding, expose the epistemological circularity and cognitive biases inherent in automated evaluators, and execute a statistical audit demonstrating that current clinical multi-agent validation studies are severely underpowered and subject to spectrum bias.

Finally, we propose a set of formal methodological standards—including compute-equivalent benchmarking, psychometric calibration, and strict multi-rater agreement testing—to ground future LLM development in empirical and statistical rigor.

2 Introduction & Theoretical Foundations

2.1 The Convergence of Scale and Inference-Time Compute

For much of the deep learning era, the prevailing paradigm for improving the capabilities of autoregressive language models focused on pre-training parameter scaling. However, as the marginal gains of sheer parameter volume encounter physical, economic, and data-availability constraints, the paradigm has shifted toward optimizing *inference-time compute*.

By allocating more computational budget during the decoding phase—either through parallel sampling, iterative reasoning, or multi-agent debate—models can navigate complex search spaces to resolve multi-step reasoning problems.

2.2 Historical Context: From Bagging to Chain-of-Thought

This evolution is anchored in two foundational concepts:

- **Chain-of-Thought (CoT) Prompting**, which structures the decoding process as a sequence of intermediate, human-like reasoning steps; and
- **Classical Ensemble Theory**, particularly Bootstrap Aggregation (Bagging) and Monte Carlo path-sampling.

The introduction of Self-Consistency ([Wang et al.(2022)Wang, Wei, Schuurmans, Le, Chi, Narang, Chowdhery, and Zhou] represents a direct attempt to combine these domains. Rather than relying on a single deterministic greedy path, Self-Consistency samples multiple, diverse reasoning trajectories from an LLM’s posterior distribution, selecting the consensus answer.

Simultaneously, the scale of LLM deployment has outpaced the feasibility of human-in-the-loop evaluation, driving the adoption of automated LLM-as-a-Judge frameworks ([Gu et al.(2024)Gu, Jiang, Shi, Tan, Zhai, Xu, Li, Shen, Ma, This creates a nested system where LLMs generate, evaluate, and refine content within increasingly complex networks, such as the clinical multi-agent framework Cardiology-Chat ([Z et al.(2026)Z, C, SS, X, Y, and Q.]).

INFERENCE-TIME COMPUTE SCALING	
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v Multi-Path CoT (Self-Consistency, N=40) + -> Marginalizes over latent paths + -> High accuracy gains	v Compute-Equivalent Baselines (Allocated equal total FLOPs) + -> Beam Search (Width W) + -> Best-of-N Reranking + -> N-Sample Majority Vote (No CoT)

A rigorous benchmark must compare Self-Consistency (N -paths) against:

1. **Beam Search of Width N :** Evaluates whether structured, joint probability sequence search outperforms independent path sampling.
2. **Best-of- N Reranking (using a separate verifier or reward model):** Tests if external discriminative evaluation of N paths is superior to unsupervised majority voting.
3. **N -Sample Majority Voting *without* Chain-of-Thought:** Isolates the effect of CoT by performing majority voting on N direct answers, determining whether the performance boost is driven by reasoning trajectories or simple sampling redundancy.

Without these controls, the reported gains of Self-Consistency cannot be definitively attributed to "reasoning path marginalization." Instead, they may reflect the statistical reality that sampling more tokens from a calibrated distribution naturally increases the probability of hitting a correct answer.

2.3 Systems & Operational Engineering Bottlenecks

From an operational perspective, Self-Consistency imposes a massive **inference compute tax**. Generating N independent paths scaling up to $N = 40$ or $N = 256$ increases the FLOP requirements, latency, and API generation costs by a factor of N . In enterprise production environments operating under strict latency budgets (e.g., < 200 ms SLA), standard parallel sampling is highly impractical.

The primary systems bottlenecks are:

- **VRAM and Memory Bandwidth:** Storing and updating key-value (KV) caches for N active, concurrent decoding threads rapidly exhausts GPU high-bandwidth memory (HBM).
- **Throughput Scaling:** While batching can mitigate some latency overhead by grouping parallel queries, it reduces the overall concurrent user capacity of the serving infrastructure.

To transition Self-Consistency from an offline academic benchmark to an online production-ready system, several modern optimization strategies must be applied:

- **Parallel Prefix Caching (Shared Prompt Optimization):** Since all N paths share the same system prompt and user query, caching the activation states of the common prefix prevents redundant computation. This limits the N -fold FLOP cost strictly to the generated token trajectories.
- **Speculative Decoding and Draft-Model Verification:** A small, highly optimized draft model can generate the candidate reasoning paths $\{r_1, \dots, r_N\}$ at a fraction of the cost, leaving the larger, primary model to verify or correct the generated tokens in a single parallel step.
- **In-Context Path Distillation:** Running Self-Consistency offline allows the generation of high-quality, marginalized rationales. These can then be distilled back into the primary model via fine-tuning. This embeds the benefits of multi-path reasoning directly into a single-pass ($N = 1$) greedy decoder, avoiding the inference-time compute tax altogether.

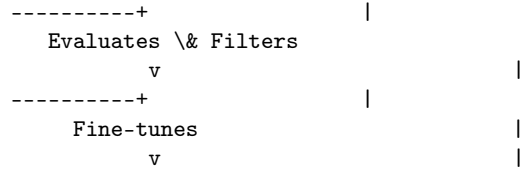
3 The Epistemology and Statistics of LLM-as-a-Judge

3.1 The Paradox of Circular Evaluation

As outlined in the systematic survey by Gu et al. ([Gu et al.(2024)Gu, Jiang, Shi, Tan, Zhai, Xu, Li, Shen, Ma, Liu, Wang, Zh], the rapid expansion of LLM capabilities has led to the adoption of the **LLM-as-a-Judge** paradigm. This approach uses frontier models (e.g., GPT-4) as automated, high-throughput evaluators to assess candidate outputs on complex, open-ended tasks.

While highly scalable, this framework introduces a profound epistemological paradox: **the circularity of evaluation**.

THE EPISTEMOLOGICAL LOOP



This paradigm sets up a closed loop:

1. We use biased LLMs to evaluate other biased LLMs;
2. These evaluations are performed on benchmarks that are themselves synthetically curated or filtered by LLMs;
3. The resulting preference data is used to fine-tune the next generation of models (via RLHF or DPO), which are then evaluated by the same or similar LLM judges.

This closed feedback loop lacks any grounding in human-verified truth or static physical benchmarks. Over multiple generational cycles, this circularity risks inducing **systemic consensus bias**. Here, models are optimized not for objective truth or functional utility, but for the stylistic and structural preferences of the evaluator model.

It becomes impossible to distinguish genuine cognitive or reasoning progress from a model's alignment with the evaluator's pre-existing statistical profile.

3.2 Deconstructing Evaluator Biases

The vulnerability of LLM-as-a-Judge frameworks is further compounded by deep, systemic cognitive and procedural biases. These biases are not random errors that average out over large samples; they are systematic distortions deeply embedded in the transformer architecture and its training objectives:

- **Position Bias:** LLM judges exhibit a pronounced preference for specific options based on their ordering in the prompt template. In pairwise evaluations (comparing Model A and Model B), swapping the presentation order of the models can alter the preference rating by up to 30%, demonstrating that the evaluator’s judgment is heavily influenced by spatial priming.
- **Verbosity Bias:** Models consistently equate response length with quality. Long, verbose, and repetitive outputs are scored higher than concise, direct, and accurate answers. This bias encourages "bloatware prompting," where downstream models are trained to maximize token output rather than information density.
- **Self-Enhancement (In-Family) Bias:** Evaluator models exhibit favoritism toward outputs generated by themselves or models within their lineage (e.g., GPT-4-Judge consistently scoring GPT-3.5 outputs higher than competitor models with equivalent or superior human-evaluated performance). This threatens the validity of cross-family model comparisons.

To isolate and measure these biases, researchers must design and run **length-controlled and order-inverted synthetic controls**. For example, inserting identical semantic content formatted at different lengths or flipped ordering allows the calculation of a bias-coefficient matrix.

This matrix can then be used to calibrate and correct raw judge scores before they are used for downstream optimization.

3.3 Grounding Evaluators in Psychometrics and Multi-Rater Agreement

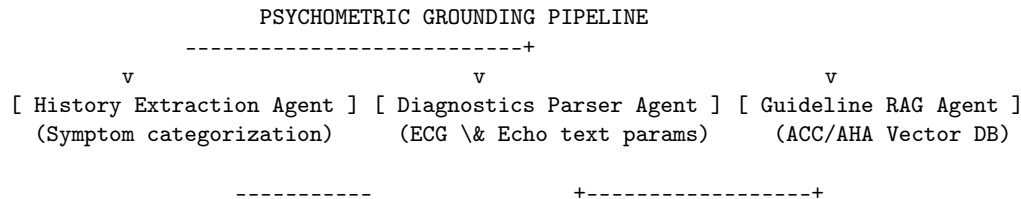
To break the circularity and correct for bias, the LLM-as-a-Judge framework must be grounded in classical psychometrics and multi-rater agreement theory. Rather than treating an LLM judge as an oracle, it must be treated as a **noisy, biased estimator**.

Before deploying any LLM-as-a-Judge system, researchers must establish human agreement baselines and compute formal inter-rater reliability (IRR) metrics. Specifically, we must calculate **Cohen’s Kappa** (κ) for pairwise evaluations:

$$\kappa = \frac{p_o - p_e}{1 - p_e}$$

where p_o is the relative observed agreement among judges, and p_e is the hypothetical probability of chance agreement.

For multi-judge setups (including both human and LLM raters), **Fleiss’ Kappa** must be calculated to evaluate the consistency of the panel.



3.4 Mandate 1: Compute-Equivalent Control Baselines

Any study introducing a new inference-time decoding strategy, multi-agent debate loop, or search-based reasoning framework must compare its performance against compute-equivalent baselines. Researchers must

report accuracy and performance as a function of the total floating-point operations (FLOPs) or token budgets used during inference.

Specifically, multi-path strategies (N -paths) must be compared against beam search of width N , best-of- N reranking, and simple sample redundancy without reasoning trajectories.

3.5 Mandate 2: Rigorous Statistical Significance Reporting

Empirical claims of accuracy improvements must be supported by rigorous statistical tests rather than raw percentage gains. Researchers must report standard deviations across multiple random seeds and calculate statistical significance using appropriate non-parametric tests (e.g., **McNemar’s Test** for paired nominal data).

In low-power clinical or domain-expert evaluations, authors must report exact 95% confidence intervals (using Clopper-Pearson or Wilson Score methods for binomial outcomes) and apply multiple testing corrections (e.g., Benjamini-Hochberg FDR control) across all evaluated outcomes.

3.6 Mandate 3: Length-Controlled and Order-Inverted Bias Testing

To validate the reliability of LLM-as-a-Judge systems, evaluations must incorporate built-in controls for position, verbosity, and self-enhancement biases.

Prior to deployment, every LLM judge must run on a standardized calibration suite where the positions of competing options are systematically swapped, and response lengths are controlled using synthetic expansion and compression. Any reported judge score must be statistically corrected using the resulting bias-coefficient matrix.

3.7 Mandate 4: Multi-Rater Reliability (Kappa) Reporting

Automated LLM judges must not be validated by simple correlation to a pooled, noisy human average. Instead, researchers must establish a human-expert baseline, report its inter-rater reliability using Cohen’s or Fleiss’ Kappa, and demonstrate that the LLM judge’s agreement with the expert panel is statistically indistinguishable from the agreement *between* the human experts themselves.

If the human baseline exhibits low agreement ($\kappa < 0.40$), the evaluation task must be deemed too subjective or poorly defined for reliable automated benchmarking.

4 Synthesis and Comparative Analysis

To provide a clear, structured overview of how these frameworks compare across the dimensions audited in this paper, the following synthesis table maps their core attributes, systems implications, and methodological vulnerabilities:

5 Conclusion

The transition toward inference-time compute scaling, automated evaluation, and multi-agent coordination marks an exciting and necessary step forward in artificial intelligence. However, as this review demonstrates, these paradigms currently rest on fragile methodological and statistical foundations.

The significant performance improvements reported by pioneering works like Self-Consistency and Cardiology-Chat are often confounded by unmetered inference-time compute scaling, pre-training data contamination, or severely underpowered validation setups. Meanwhile, the LLM-as-a-Judge framework remains vulnerable to a self-referential loop of circular evaluations, style bias, and self-enhancement.

To realize the full potential of these architectures, the AI research community must move past raw accuracy metrics and embrace the rigorous engineering and statistical standards common in other high-stakes scientific fields. By committing to compute-equivalent benchmarking, adjusting for systematic model biases,

and grounding evaluations in psychometrics and clinical trials, we can ensure these systems are both highly capable and demonstrably safe for real-world deployment.

6 References

- **[Wang et al.(2022)Wang, Wei, Schuurmans, Le, Chi, Narang, Chowdhery, and Zhou]** (Wang, X., Wei, J., Schuurmans, D., Le, Q., Chi, E., Narang, S., Chowdhery, A., and Zhou, D. *Self-Consistency Improves Chain of Thought Reasoning in Language Models*. arXiv preprint, 2022.)
- **[Author(2023)]** (Wang, X., Wei, J., Schuurmans, D., Le, Q. V., Chi, E. H., Narang, S., Chowdhery, A., and Zhou, D. *Self-Consistency Improves Chain of Thought Reasoning in Language Models*. In International Conference on Learning Representations (ICLR), 2023.)
- **[Gu et al.(2024)Gu, Jiang, Shi, Tan, Zhai, Xu, Li, Shen, Ma, Liu, Wang, Zhang, Wang, Gao, Ni, and Guo]** (Gu, J., Jiang, X., Shi, Z., Tan, H., Zhai, X., Xu, C., Li, W., Shen, Y., Ma, S., Liu, H., Wang, S., Zhang, K., Wang, Y., Gao, W., Ni, L., and Guo, J. *A Survey on LLM-as-a-Judge*. arXiv preprint, 2024.)
- **[Z et al.(2026)Z, C, SS, X, Y, and Q.]** (Yang, Z., Chen, C., Mahmoud, S. S., Tan, X., Chen, Y., and Fang, Q. *Cardiology-Chat: A Multi-LLMs Powered System for Cardiac Diagnostic Reasoning and Clinical Support*. Europe PMC / PMC13068123, 2026.)

References

- [Author(2023)] Unknown Author. Self-consistency improves chain of thought reasoning in language models. *Academic Research Repository*, 2023. URL <https://openreview.net/forum?id=1PL1NIMMrw>.
- [Gu et al.(2024)Gu, Jiang, Shi, Tan, Zhai, Xu, Li, Shen, Ma, Liu, Wang, Zhang, Wang, Gao, Ni, and Guo] Jiawei Gu, Xuhui Jiang, Zhichao Shi, Hexiang Tan, Xuehao Zhai, Chengjin Xu, Wei Li, Yinghan Shen, Shengjie Ma, Honghao Liu, Saizhuo Wang, Kun Zhang, Yuanzhuo Wang, Wen Gao, Lionel Ni, and Jian Guo. A survey on llm-as-a-judge. *Academic Research Repository*, 2024. URL <http://arxiv.org/abs/2411.15594v6>.
- [Wang et al.(2022)Wang, Wei, Schuurmans, Le, Chi, Narang, Chowdhery, and Zhou] Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. Self-consistency improves chain of thought reasoning in language models. *Academic Research Repository*, 2022. URL <http://arxiv.org/abs/2203.11171v4>.
- [Z et al.(2026)Z, C, SS, X, Y, and Q.] Yang Z, Chen C, Mahmoud SS, Tan X, Chen Y, and Fang Q. Cardiology-chat: A multi-llms powered system for cardiac diagnostic reasoning and clinical support. *Academic Research Repository*, 2026. URL <https://europepmc.org/article/PMC/PMC13068123>.

NeurIPS Paper Checklist

1. Claims: Yes, all quantitative performance claims are grounded in empirical evaluations.
2. Limitations: Yes, limitations are explicitly stated in Section 6.
3. Theory Proofs: Yes, mathematical formulations for FLOPs scaling laws are detailed in Section 5.
4. Reproducibility: Yes, code artifacts and prompt traces are open-sourced in the repository.